Alternating Series

① Show $\sum a_n$ is abs. conv.

Check: $\sum |a_n|$ is conv.

② Show $\sum a_n$ is cond. conv.

- $\sum |a_n|$ is div.
- Use AST to show $\sum a_n$ conv.

③ Show $\sum a_n$ is div.

- Use n-term test

$\lim |a_n| \neq 0 \Rightarrow \cancel{\lim \{a_n\} \neq 0}$

$\lim a_n \neq 0$

Ex. 9.6

Q28

$$\sum_{n=2}^{\infty} (-1)^{n+1} \frac{1}{n \ln n}$$

① (Integral test)

$$a_n = (-1)^{n+1} \frac{1}{n \ln n}$$

[Aim: $\sum |a_n|$ div. by Int. test]

② (AST)

$$b_n = \frac{1}{n \ln n}$$

$b_n \geq 0$

b_n eventually dec.

$b_n \geq b_{n+1}$ for some $n \geq N$

$b_n \to 0$

$\therefore \sum a_n$ conv. $\Rightarrow \sum a_n$ cond. conv.

Let $c_n = |a_n|$

$$\sum c_n = \sum_{n=2}^{\infty} \frac{1}{n \ln n} \text{ div. or conv.?}$$

Let $f : [2, \infty) \to \mathbb{R}$ be defined by

decreasing

$$f(x) = \frac{1}{x \ln x} \text{ for } x \geq 2.$$

nonnegative Continuous!

$[f(n) = c_n$ for all integers $n \geq 2]$

$$\int_2^{\boxed{\infty}} \frac{1}{x \ln x}\, dx = \lim_{\boxed{k} \to \infty} \int_2^{\boxed{k}} \frac{1}{x \ln x}\, dx$$

$$= \cdots$$

$$= \infty$$

Clearly f ~~(x)~~ is decreasing and continuous for $x \geq 2$

$\because$ $x \ln x$ is increasing for $x \geq 2$.

Clearly $f(x) \geq 0$ for $x \geq 2$.

$$\therefore \sum_{n=2}^{\infty} c_n = \sum_{n=2}^{\infty} \frac{1}{n \ln n} \text{ div.}$$

by Int. test.

$$\boxed{\therefore \sum_{n=2}^{\infty} |a_n| \text{ div.}}$$

② (AST)

Clearly $b_n = \dfrac{1}{n\ln n} \geqslant 0$

for $n \geqslant 2$.

Since $n \ln n$ is increasing on $n \geqslant 2$,

so $b_n = \dfrac{1}{n \ln n}$ is decreasing on $n \geqslant 2$.

$\therefore b_n \geqslant b_{n+1} \ \forall n \geqslant 2$.

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{1}{n\ln n} = 0 \quad \boxed{\neq \frac{1}{\infty}}$$

$\dfrac{1}{n\ln n} \leq \underset{\substack{\downarrow \\ 0}}{\dfrac{1}{n}}$ $\quad \left[\because \lim_{n\to\infty} n\ln n = \infty\right]$

Int. Test

Let $f(x) = \cdots$

Clearly $f(n) = a_n$ for $n \geqslant \ldots$

$$\int_{\ldots}^{\infty} f(x)\,dx$$

$$= \lim_{c\to\infty} \int_{\ldots}^{c} f(x)\,dx$$

$\vdots$

$= \square$